EMBEDDED FIGURES

Finding a Figure Hidden in a Complex Figure

OAVS TGT | Reasoning (Additional topic from the “etc.” in the syllabus) | Topic 22

Syllabus line (Reasoning, 30 Marks): The verbal and non-verbal topics listed in the syllabus end with “etc.”. This topic is not named in that line, but it is asked in the papers: embedded-figure questions appear in Social Studies 2023 (Q18 and Q22).

Built from: OAVS TGT English 2019, English 2022, Mathematics 2024, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2). Two embedded-figure questions appear (Social Studies 2023 Q18, Q22). Both are solved with their original figures. The practice questions were generated or checked by program so that each has exactly one correct option.

What is inside

Ch.	Title	What you get
1	PYQ Analysis	The 2 PYQs and their types
2	Foundation	Search method for hidden figures
3	Solved PYQs	2 PYQs with original figures
4	Shortcuts, Traps and Scope	Speed methods and mistakes
5	Practice Set	6 original questions with a verified key
6	Quick Revision	One-page cheat sheet

CHAPTER 1

PYQ Analysis

Two questions, one paper

Paper	Q.	Question	Type	Ans
Social Studies 2023	18	Select the option in which the given figure is embedded (rotation not allowed)	Find the option that contains it	4
Social Studies 2023	22	Select the option that is NOT embedded in the given figure (rotation not allowed)	Find the option that is not inside	3
English 2019 / English 2022 / Mathematics 2024 / Science 2024	–	none		

PYQ INSIGHT: Both PYQs are from the **same paper**, one asking the figure to be found in an option and one asking which option is not contained in the big figure. Both say **rotation is NOT allowed**, so the hidden figure must appear in the same orientation (it can be in any position).

Pattern	PYQs	Priority
Find the option that contains the figure	1	★★★ Must
Find the option that is NOT contained in the big figure	1	★★★ Must
Count the number of hidden figures	0	★ Possible

CHAPTER 2

Foundation

How to find a hidden figure

2.1 Two types

Type	Task	Method
Embedded	Pick the option that contains the question figure	Look for the most distinctive part of the figure in each option; reject options that lack it
NOT embedded	Pick the small figure that cannot be found inside the big figure	Check each small figure against the big one; the odd one has a line or angle that does not exist

2.2 Method

Step	Do this
1	Pick the **most distinctive feature** of the small figure: a curve, a closed shape, an unusual angle, the longest line.
2	Find that feature in each option (or in the big figure) and check its position relative to the rest.
3	Trace the whole figure from that spot; every line must be present, in the same direction (no rotation).
4	Reject options where one line is missing, extra lines crossing the figure do not matter (they are allowed), but missing lines do.

NOTE: In an embedded question, **extra lines are allowed** in the big figure; only **missing lines** make an option wrong.

SHORTCUT: Count the lines of each direction (horizontal, vertical, slanting) in the small figure; an option with fewer lines of any direction cannot contain it.

CHAPTER 3

Solved PYQs

Both questions with original figures

Social Studies 2023 Q18

Select the option in which the given figure is embedded (rotation is NOT allowed).



Figure 1 2 3 4 ✓

1. The figure has a small circle on top of a vertical stem, with two small squares attached left and right, and a slanted tail at the bottom.
2. Use the circle with the stem and the two squares as the key feature: they must appear together in the same orientation.
3. Options 1, 2 and 3 do not contain the circle, stem and both squares in this arrangement; option 4 does, with the tail at the bottom.
4. The scan is small, so the elimination of options 1 to 3 is read from the figures. The official key is option 4.
Answer: Option 4

Social Studies 2023 Q22

Select the option that is NOT embedded in the given figure (rotation is NOT allowed).

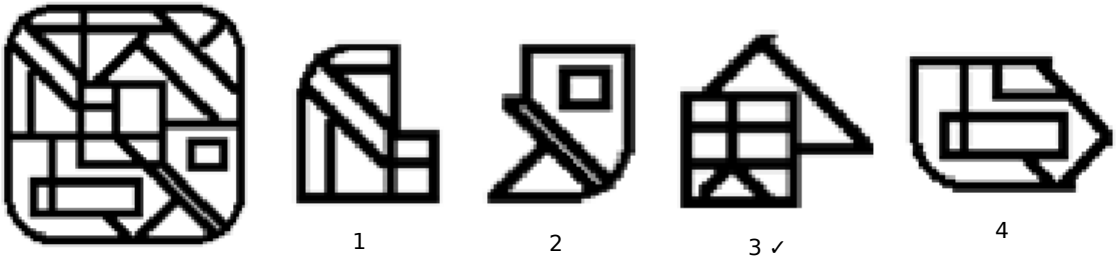


Figure 1 2 3 ✓ 4

1. Check each small figure against the large rounded-square figure: find a distinctive part (a diagonal with a small square, an inner rectangle) and see whether it exists in the same position.
2. Options 1, 2 and 4 can be found inside the large figure; each has a diagonal or rectangle that appears in the same direction.
3. Option 3 (the house-shaped figure with a grid and a triangular roof) cannot be found: its roof and grid lines do not occur together in the big figure.
4. The scan is small, so the elimination is read from the figures. The official key is option 3.
Answer: Option 3

CHAPTER 4

Shortcuts, Traps and Scope

Speed methods and mistakes

- **Start from the odd part:** a curve or a closed shape is easier to find than a straight line.
- **Direction counts:** the small figure may not be rotated, so slants (“\” or “/”) must match.
- **Extra lines are fine** in the big figure; missing lines are not.
- **Eliminate fast:** reject any option that lacks one line of the figure.
- **Time:** 45 seconds per question.

Trap	How to avoid it
Rotating the figure to fit	Rotation is not allowed; check the slant of every line.
Ignoring size	The hidden figure has the same size as the question figure.
Rejecting an option because of extra lines	Extra lines are allowed in the big figure.
Missing one small line	Check every line, not just the main shape.

Scope

Related PYQs / topics	Where
Completion of a pattern (the missing quadrant)	Completion of Pattern (Topic 15)
Figure formation from pieces	Figure Formation (earlier topic)

CHAPTER 5

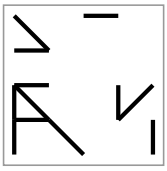
Practice Set

6 original questions; answers on the next page

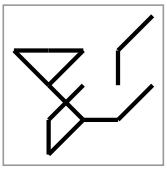
A. Which option contains the figure?

Q1. In which option is the figure shown below embedded? (The figure must be in the same position and not rotated.)

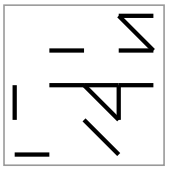




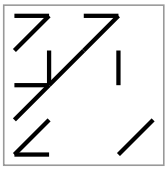
A



B



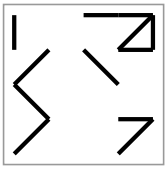
C



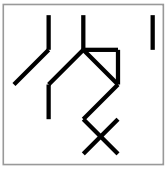
D

Q2. In which option is the figure shown below embedded? (The figure must be in the same position and not rotated.)





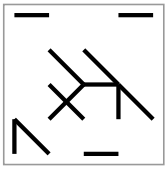
A



B



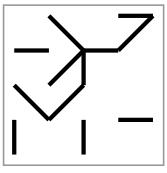
C



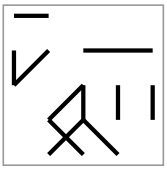
D

Q3. In which option is the figure shown below embedded? (The figure must be in the same position and not rotated.)





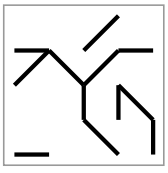
A



B

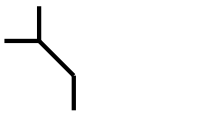


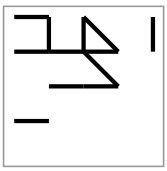
C



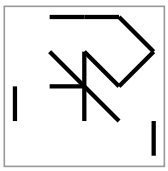
D

Q4. In which option is the figure shown below embedded? (The figure must be in the same position and not rotated.)

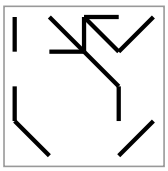




A



B



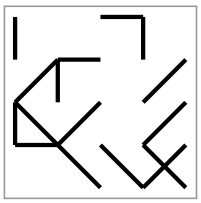
C



D

B. Which option is NOT embedded?

Q5. Select the option that is NOT embedded in the given figure (no rotation).

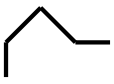




A



B

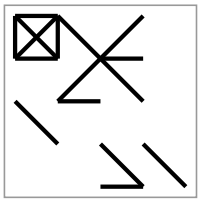


C



D

Q6. Select the option that is NOT embedded in the given figure (no rotation).





A



B



C



D

Answer key

Q	Answer	Reason
1	B	Option B contains every line of the given figure in the same orientation (checked by program). Each of the others lacks at least one of its lines.
2	D	Option D contains every line of the given figure in the same orientation (checked by program). Each of the others lacks at least one of its lines.
3	A	Option A contains every line of the given figure in the same orientation (checked by program). Each of the others lacks at least one of its lines.
4	C	Option C contains every line of the given figure in the same orientation (checked by program). Each of the others lacks at least one of its lines.
5	C	Option C needs a line that does not appear in the large figure in that orientation; the other three can all be found in it (checked by program).
6	A	Option A needs a line that does not appear in the large figure in that orientation; the other three can all be found in it (checked by program).

CHAPTER 6

Quick Revision

One page for the last day

Question type	Method
Embedded	Find the most distinctive part; check every line; extra lines allowed
NOT embedded	The option with a line that does not exist in the big figure
Rotation not allowed	Check slants and orientation
Same size	The hidden figure is not enlarged or reduced

PYQ RULE: Both PYQs are from Social Studies 2023: find the option that contains the figure (Q18), and the option that is not contained (Q22). Rotation is not allowed.

ROUTINE: Exam-day routine: find the most distinctive part; trace the whole figure; check every line in the same orientation; eliminate. 45 seconds per question.